Interclass Separability Factorization for Hyperspectral Anti-Interference Unmixing

Chunzhi Li, Yangyang Zhang, Xiaohua Chen, Huimeng Zheng, and Jiaxin Li, *Member, IEEE*

***Abstract*—In the field of hyperspectral unmixing (HU), to decompose mixed pixels into pure substances and their corresponding fractional abundances, the (hyper)graph constrained (*D*) nonnegative matrix factorization (NMF) is typically applied under the guidance of the spectral information, and as a result, the unmixing accuracy is highly dependent on the construction method of the (hyper)graph and how the spectral information is exploited. In addition, the (hyper)graph constrained (*D*)(NMF) models are usually trapped in the problem of over-smoothness, that is, the pixels belonging to different substances and located on the edges are inevitably distorted. In this article, a weighted NMF is constructed based on the spectral similarity between the bands, which is exploited from the physical characteristics of the hyperspectral image (HSI) and the corresponding interferences, respectively. Subsequently, a multihigher-order graph is pursued and introduced to exploit the local smoothness of the abundances. This approach ensures that the edge information of each abundance map is expected to be preserved, and interclass pixels can be successfully separated. Finally, an effective unmixing model called multigraph regularized nonnegative matrix weighted factorization (MGNMWF) is constructed. The proposed model is experimented on both synthetic and real datasets. The experimental results show that the proposed model outperforms both shallow and deep unmixing models.**

***Index Terms*—Hyperspectral unmixing (HU), interclass separability, multihigher-order graph regularization, spectral similarity.**

I. Introduction

With the development of the hyperspectral imaging technology, the acquired hyperspectral image (HSI) has found applications in an increasing number of fields [1], [2], [3]. Unlike the traditional natural images, the HSI possesses rich spectral information, and the dimensionality increases greatly with the development of hyperspectral imaging technology [4]. However, the increase in spectral information is accompanied by the low spatial resolution, leading to the presence of mixed pixels in HSIs [5]. Since the HU technology aims to decompose the mixed pixels into pure substance spectra and corresponding fractions, that is, endmembers and abundances, the hyperspectral unmixing (HU) technology becomes indispensable to improve the HSI application. Most existing unmixing methods are based on either the linear mixing model (LMM) or the nonlinear mixing model (NLMM). Due to their simplicity and clear physical interpretation, LMM-based methods have become more prevalent than NLMM-based methods. Based on the LMM, the HU methods can be coarsely classified into the following categories: geometrical methods, sparse regression methods, deep learning-based methods, statistical methods, and so on. Specifically, geometrical methods embed the data points into the simplex of the feature spaces [6], sparse regression methods generally require the spectral library or known endmembers [7], and deep learning-based methods excel at capturing the fine features from the multilayered structures [8]. In contrast to these three categories of LMM-based methods, the statistical methods have attracted more attention, particularly, the nonnegative matrix factorization (NMF)-based methods are a category of canonical statistical methods. NMF-based statistical unmixing methods are widely used due to their interpretability and ability to derive results without any prior information.

Beyond the classification approaches mentioned above, unmixing models have also been broadly classified into physics-based, data-driven models, as well as an integrated model that combines physics-based and data-driven approaches [9]. Similar to NMF-based unmixing methods, the physics-based models utilize statistical properties to represent the spectral characteristics and interactions between different substances present in the imaged scene. Along the line of physics-based unmixing methods, some recent representative methods are summarized as follows. Qian et al. [10] replaced the L_1 norm with the $L_{1/2}$ norm and proposed $L_{1/2}$–NMF to achieve sparser unmixing results. Based on the $L_{1/2}$–NMF, Feng et al. [11] introduced total variational (TV) regularization to suppress the noise and improve the smoothness of the unmixing results. To improve the unmixing performance from both spectral and spatial perspectives, Yang et al. [12] proposed a spectral–spatial anti-interference unmixing algorithm. The anti-interference algorithm employed superpixels to impose homogeneous constraints on the pixels within the

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neighborhood, and the spectral correlation was exploited to separate the interferences from the main information. Inspired by graph learning, Qu et al. [13] introduced the second-order nearest neighbor relation to overcome the deficiency of the first-order graph, and a high-order graph regularized NMF was presented for HU. Considering that the HSI is usually unevenly distributed along the spectral range, [14] presented a functional factorization model via incorporating the latent neural representations to exploit the endmembers. To reduce the correlation between the pure substances and to exploit the spatial structure, Song et al. [15] introduced the mutual incoherence and relative TV-constrained NMF for the task of HU. To reconstruct the abundance details and overcome the noise, Zou et al. proposed a graph Laplacian weighted HU method [22]. To implement the graph Laplacian weighted method, Zou et al. introduced the simple iterative convergence (SLIC) technique to segment the HSI [22]. Unlike the aforementioned physics-driven methods, data-driven unmixing methods typically use advanced deep learning methods. However, as in unmixing using the autoencoder network [16], their theoretical underpinnings remain elusive. Specifically, physically uninterpretable deep learning models tend to disregard abundance similarity. Consequently, deep learning-based models are less effective in complex scenarios due to their inadequate use of spatial information. Furthermore, these methods typically struggle to address spectral variability issues due to their limited ability to exploit critical spectral characteristics. Recently, several integrated models combining physics-based and data-driven approaches have been proposed. For instance, Zhang et al. [17] introduced a fully linear mixture model and designed a spectral variability-aware cascaded autoencoder for HU tasks. This approach autonomously handles diverse spectral variations without requiring prior information. It considers variations in reflectance caused by scene geometry and lighting conditions. Liu et al. [18] integrated linear and nonlinear models within a deep learning framework to balance robustness and accuracy. Li et al. [19] applied noise filter guidance constraints to a deep learning-based NMF model. However, it is worth noting that these methods require extensive empirical tuning of multiple hyperparameters, which demonstrates the significant complexity of model optimization and parameter adjustment.

Obviously, the emerging physics-based methods have concentrated on HU by addressing nonuniformly distributed spectral domains and/or spatial structural details. However, the existing HU methods are insufficient to exploit the relationship of the pixels within the neighborhood, resulting in the boundary between the interclass substances being smoothed, and the different substances are not separated. For example, [13] applies graph regularization globally without considering homogeneity. Although homogeneity was taken into account method [12], the SLIC used ignored the edge information. In addition, existing HU methods only partially considered the physical meaning of the spectra and typically disregarded the similarity between the bands. Although algorithm [13] selects features for bands using adaptive weights, it ignores the correlation between bands. To this end, we exploit the similarity between the bands by capturing their relationship in the reflectance data of a given pixel at different frequency bands while performing feature selection on the bands, and then we fully exploit the local smoothness of an abundance by a novel segmentation-driven local graph regularization method. The main contributions of our work are listed below.

1) We propose a novel spectral affinity constraint for the weighted factorization model that preserves interband correlations during endmember estimation. Our approach enforces band-to-band similarity to ensure the resulting spectral signatures are physically meaningful and align with the physics of hyperspectral imaging.
2) We present a novel boundary refinement method that uses localized, high-order graph modeling to precisely delineate interclass boundaries while maintaining regional smoothness. Unlike existing global graph regularizers, our technique provides a more physically meaningful, segmentation-guided graph constraint by adaptively coupling spectral signatures with spatial relationships.
3) The proposed unmixing method systematically aggregates the spectral similarity and local smoothness of the spatial subspace. Extensive experiments and vigorous results demonstrate the superiority of the proposed unmixing method.

II. Related Works

In this section, the related basic theories of HU are introduced. Specifically, the mathematical formulas of the LMM, NMF, and sparse NMF are illustrated, respectively.

A. Linear Mixing Model

Generally, the LMM assumes that a mixed pixel can be approximated as a linear combination of several endmembers. Let $X = [x_1, x_2, \ldots, x_N] \in \mathbb{R}^{b\times N}$ denote an HSI data, then the LMM can be written mathematically as

$$X = \mathrm{UV} + E \tag{1}$$

where $U \in \mathbb{R}^{b\times P}$ denotes the endmember matrix, $V \in \mathbb{R}^{P\times N}$ denotes the abundance matrix, $E \in \mathbb{R}^{b\times N}$ is the noise matrix, and b, P, and N, respectively, denote the number of spectral bands, endmembers, and pixels. In particular, both the endmember matrix U and the abundance matrix V are nonnegative, and the abundance matrix V should satisfy the sum-to-one constraint [15], that is, $u_{ij}, v_{ij} \geq 0$, $\sum_{i=1}^{P} v_{ij} = 1$, where u_{ij} and v_{ij}, respectively, denote the elements of the endmember and abundance matrices.

B. Nonnegative Matrix Factorization

Since our proposed unmixing method is based on the NMF, we illustrate the NMF in detail in this section.

The NMF aims to decompose a nonnegative matrix into two nonnegative low-rank matrices, that is, the product of the two decomposed nonnegative matrices approximates the original matrix. The mathematical formula can be written as follows:

$$X \approx \mathrm{UV}. \tag{2}$$

It is worth noting that the Euclidean distance is commonly employed for the task of HU to measure the dissimilarity

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between the HSI matrix X and the decomposed results UV. The NMF-based unmixing model is written as follows:

$$\min_{U\geq 0,V\geq 0} \frac{1}{2}\|X-\mathrm{UV}\|_F^2 \tag{3}$$

where $\|\cdot\|_F$ demotes the Frobenius norm.

C. Regularized NMF

Since the objective function of the common NMF is nonconvex, that is, the solution is not unique, to this end, the sparse constraint is usually employed to regularize the NMF. For example, $L_{1/2}$–NMF exploited the $L_{1/2}$ regularizer and added the regularizer into the NMF model [10], and the sparse regularized NMF model is written as follows:

$$\min_{U\geq 0,V\geq 0} \frac{1}{2}\|X-\mathrm{UV}\|_F^2+\mu\|V\|_{\frac{1}{2}} \tag{4}$$

where $\|V\|_{(1/2)}$ is defined as $\sum_{i,j=1}^{P,N}(v_{ij})^{(1/2)}$ [10] and μ is the tradeoff parameter.

Unlike the sparse regularized NMF, Qu et al. [13] considered that the first-order nearest neighbor is insufficient to represent the shared pixel structure within a neighborhood. To address this limitation, they combined the second-order nearest neighbor with the first-order nearest neighbor to overcome the deficiency of the first-order nearest neighbor. The unmixing model advocated in the literature [13] was written as follows:

$$\min_{U\geq 0,V\geq 0} \frac{1}{2}\|\Lambda(X-\mathrm{UV})\|_F^2+\mu\|V\|_{\frac{1}{2}}+\frac{\lambda}{2}\mathrm{Tr}\left(\mathrm{VL}_h V^{\mathrm{T}}\right) \tag{5}$$

where the matrix $\Lambda\in\mathbb{R}^{b\times b}$ is a nonnegative diagonal matrix, which is called an adaptive weight matrix in the literature [13]. And the matrix $L_h\in\mathbb{R}^{N\times N}$ denotes the high-order Laplace matrix. $(\cdot)^{\mathrm{T}}$ computes the transpose of any matrix or vector.

III. Methodology

This section details the motivation, justifications, and technologies of our proposed method. The research in [13] argued that the features of an HSI vary greatly in different bands, that is, the endmember features contained across different bands are unevenly distributed. According to (5), Qu et al. [13] introduced a weighted parameter Λ to capture the tiny features. It is worth noting that the weighted parameter Λ is iteratively updated by solving the problem $\min_{\Lambda\geq 0}(1/2)\|\Lambda(X-\mathrm{UV})\|_F^2$, that is, the weighted parameter Λ will be a large constant when the construction UV is tiny. Conversely, if the construction UV is large, the parameter Λ will be a tiny constant. In this way, the weighted parameter Λ regulates the imbalanced features by capturing the tiny features and suppressing the large features, that is, the similarity between different bands of an endmember and the isometric properties within the neighborhood of different abundances are not specifically considered, but the weighted parameter Λ generally regulates the products of the endmember matrix U and the abundance matrix V. The degradation of endmembers/abundances will propagate further into abundance/endmember; to this end, the literature [12] introduced two weighted parameters and a superpixel-guided regularizer to balance the relationship between the endmembers and abundances. However, a new trouble arises at this point. The effectiveness of the superpixel-guided regularizer is highly dependent on the segmentation/clustering method and the design of the regularizer structure.

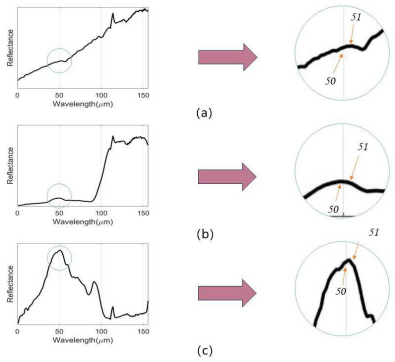


Fig. 1. Spectra of (a) soil, (b) tree, and (c) water reflectance values vary with wavelength.

In this article, we have systematically considered the properties of both spectra and abundances. We then provide a detailed description of our proposed method.

A. Spectral Similarity

The problem of uneven distribution in spectral space exists in unmixing work [13], [14]. Therefore, we introduce spectral structure into the NMF model to improve physical plausibility. However, existing methods typically only consider tiny frequency values or the smoothness of the spectrum [14]. These methods rarely consider both simultaneously, resulting in extracted end-element spectra that are ideal but inconsistent with real-world spectra.

To illustrate the spectral characteristics, three representative real spectra are shown in Fig. 1. It is obvious that the reflectance values vary greatly across the spectral domain of each substance. Notably, the values of the neighboring bands, which are circled and highlighted in Fig. 1, are highly similar to each other. Therefore, according to the spectral properties, to fully exploit the complete information from the spectral domain, the unmixing model should guarantee the continuity of the spectra and prevent the tiny reflectance contribution from being neglected. To address this issue, we introduce a weighting parameter Λ to guide the factorization of HSI data X, and design a correlation coefficient to measure the similarity between different frequency bands. Specifically, a simple heat function [20] is employed to calculate the relationship between any two bands, that is,

$$w_{i,j}=e^{-\frac{\|x_i-x_j\|}{\sigma_1}} \tag{6}$$

where $w_{i,j}$ denotes the correlation coefficient of the ith band and the jth band and x_i and x_j denote the reflectance values of the ith and jth bands, respectively. σ denotes the standard deviation, which, for any two different spectral bands x_i and x_j, σ can be calculated as follows: $\sigma=\left((1/b(b-1))\sum_{i=1}^{b}\sum_{j=1}^{b}(a_{ij}-\overline{a_{ij}})^2\right)^{1/2}$, where a_{ij} and $\overline{a_{ij}}$ are each calculated as $\|x_i-x_j\|_2$ and $\overline{a_{ij}}=\left(\sum_{i=1}^{b}\sum_{j=1}^{b}a_{ij}/b(b-1)\right)$. According to formula (6), if the

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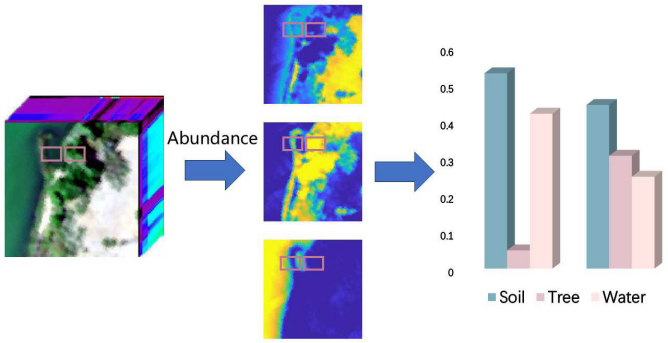


Fig. 2. Ratio of different substances for two example neighboring pixels. Specifically, two adjacent pixels, each belonging to two boxed rectangles, have a different ratio for different substances.

values of the ith band are more similar to the jth band, a larger correlation coefficient vector $w_{i,j}$ will be obtained. Conversely, if two bands are more dissimilar, a smaller correlation coefficient vector $w_{i,j}$ will be obtained. Therefore, a spectral similarity regularizer is developed to maintain the smoothness and prevent distortion, as follows:

$$\sum_{i=1}^{b}\sum_{j=1}^{b} w_{i,j}\left\|u_i - u_j\right\|^2 = \mathrm{Tr}\left(U^{\mathrm{T}}\mathrm{LU}\right) \tag{7}$$

where u_i and u_j are the endmember vectors corresponding to the ith and jth bands, respectively. $\mathrm{Tr}(\cdot)$ means to calculate the trace. L is derived from the formula (6), that is,

$$L = D - W. \tag{8}$$

And D is a diagonal matrix, calculated as $D_{ii} = \sum_{j=1}^{N} W_{ij}$.

As discussed above, we use the spectral similarity regularizer in the weighted factorization model to account for both minute frequency values and smoothness of the spectrum.

B. Interclass Separability

Current unmixing methods often fail to balance the relationship between spatial sparsity and smoothness. This results in adjacent pixels at the boundary being blindly assigned to the same class. To reverse this, the local graph regularizer must be applied to an advanced unmixing model to achieve interclass separability. Generally, two closer pixels in the HSI often have more similar abundances, so a first- or second-order (hyper)graph was employed to smooth the abundances [13], [21]. However, neighboring pixels at the boundary do not always adhere to this general rule, as in the case of Fig. 2. To highlight the heterogeneity of materials between adjacent pixels, two small regions (effectively representing single pixels) are outlined. Despite their proximity, these pixels exhibit markedly distinct abundance compositions. Specifically, according to Fig. 2, the former boxed pixel consists of more than 55% soil, 40% water, and less than 1% tree. In comparison, the latter boxed pixel has more than 45% soil, 25% water, and 30% tree. This significant variation in endmember fractions shows the abrupt spatial heterogeneity present at material boundaries. However, traditional graph constraint is based on the assumption that the neighboring pixels are composed of similar materials [12]; this assumption conflicts with the distribution in the real scene, as a result, interclass pixels at the boundary are inevitably not separated for the smoothness constraint. To overcome the shortcomings of the assumption and to capture more abundant local details, superpixels have been increasingly used in recent unmixing studies [22]. As the SLIC algorithm is a well-known and efficient segmentation method in the field of natural image processing [22], it is regularly employed to segment the HSI. It is worth noting that the SLIC generally exploits both spatial and color information to achieve the segmentation task, while the HSI is very different from the natural image, as the HSI has rich spectral information, while the natural image has only red, green, and blue channels. Considering the difference between the natural image and the HSI, the literature [22] improved the traditional SLIC by assigning the pixels to the nearest seed point with both the spectral and spatial information. However, it was found that the improved SLIC in the literature [22] only introduced the spectral information via the Euclidean distance.

In this article, we propose a more appropriate segmentation method to exploit the superpixels in the HSI. In this way, a higher-order graph regularizer can be enforced on each appropriate superpixel to avoid the phenomena of over-smoothness and guarantee the accuracy of the abundances. In particular, the accuracy of the abundance will be guaranteed by successfully separating the interclass pixels at the boundary. Specifically, both the shape and the reflectance values are considered to improve the accuracy of the obtained superpixels. The traditional Euclidean distance is good at measuring the similarity of reflectance values between different spectra, but it is insufficient to measure the shape of the spectra. Spectral angle distance (SAD) has been used widely to evaluate the accuracy of the extracted endmembers, as it is excellent at measuring both the shape and the reflectance values [23]. In contrast to the Euclidean distance, the SAD is inferior in measuring the spectral reflectance contributions and is sensitive to noise. Therefore, to fully exploit the spectral information for pursuing more accurate superpixels, we introduce both the Euclidean distance and the SAD into a segmentation method. Let the HSI cube be Y. Without loss of generality, the initial seed points are set as in the traditional SLIC method. Then the spatial and spectral Euclidean distances and the SAD are calculated, respectively, as follows:

$$c_{p_{ij}} = \sqrt{\left(\alpha_i - \alpha_j\right)^2 + \left(\beta_i - \beta_j\right)^2} \tag{9}$$

$$c_{\mathrm{se}_{ij}} = \left\|\gamma_i - \gamma_j\right\|_2^2 \tag{10}$$

$$c_{\mathrm{sa}_{ij}} = \arccos\left(\frac{\gamma_i^{\mathrm{T}}\gamma_j}{\left\|\gamma_i^{\mathrm{T}}\right\|\left\|\gamma_j\right\|}\right) \tag{11}$$

where $c_{p_{ij}}, c_{\mathrm{se}_{ij}}$, and $c_{\mathrm{sa}_{ij}}$ denote the spatial, spectral Euclidean distance, and SAD, respectively. α_i and β_i are the coordinate values of the horizontal axis and vertical axis for the ith pixel in the spatial spaces, respectively, and α_j and β_j are the coordinate values of the horizontal axis and vertical axis for the jth seed point in the spatial spaces, respectively. γ_i and γ_j denote the ith spectral vector and the jth seed spectral vector, respectively. Therefore, the total distance c_{ij} for optimizing the

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superpixels with the consideration of both spatial and spectral information can be calculated as $c_{ij} = (c_{p_{ij}}^2 + c_{\mathrm{se}_{ij}}^2 + c_{\mathrm{sa}_{ij}}^2)^{1/2}$. However, it is clear that the total distance c_{ij} equally weights three kinds of distances $c_{p_{ij}}$, $c_{se_{ij}}$, and $c_{\mathrm{sa}_{ij}}$, which means that the edge information is not fully exploited and the pixels at the edge are most likely to be inaccurately segmented. Taking two neighboring points y_i and y_z in the HSI cube Y as an example, if these two adjacent pixels belong to different classes and are located at their respective edges, these two neighboring points usually tend to have a minimized spatial Euclidean distance between the same seed point and are misclassified into the same class. To solve this problem, it is necessary to use the edge information to improve the total distance c_{ij}. Since the spectral information is beneficial to highlight the pixels at the edge, the pixels at the edge contain more spectral information than the pixels at the nonedge. Moreover, if the pixels are at the edge, the corresponding spectra are at the edge, while the reason for incorrect segmentation for the neighboring pixels is mainly the high similarity between the neighboring pixels.

To avoid the incorrect segmentation for the neighboring pixels, the gradient of the spatial space is exploited using a regular edge detection operator, the Sobel operator. Since the Sobel operator is professional at the task of detecting the edge of any 2-D image [24], we detect the edge of the spatial space in the HSI cube, and the edge information is effectively represented by the gradients, so we calculate the horizontal and vertical gradients by multiplying the Sobel operator with the spatial image at the ith spectrum along the horizontal and vertical directions, respectively. Let the horizontal and vertical gradients at the ith spectrum be G_{α_i} and G_{β_i}, the total spatial gradient along all the spectral bands can be calculated as follows:

$$G = \sum_{i=1}^{b} \left(\left| G_{\alpha_i} \right| + \left| G_{\beta_i} \right| \right). \tag{12}$$

where $|\cdot|$ calculates the absolute value. The normalized total gradient $\widetilde{G}$ is calculated as $\widetilde{G} = (G/G_{\max})$, where $G_{\max}$ is the maximum spatial gradient value. The normalized gradient, $\widetilde{G}$, then serves as a weighting factor to adjust the spectral distance measures $c_{\mathrm{se}_{ij}}$ and $c_{\mathrm{sa}_{ij}}$. However, the scale differences inherent in these three distance metrics may disproportionately emphasize larger-valued components, potentially masking subtle yet important variations. To address this issue, normalization is performed to eliminate dimensional inconsistencies. The final composite distance metric for superpixel updating is therefore formulated as follows:

$$c_{ij} = \sqrt{\widetilde{c_{p_{ij}}}^2 + \widetilde{G}\left(\widetilde{c_{\mathrm{se}_{ij}}}^2 + \widetilde{c_{\mathrm{sa}_{ij}}}^2\right)}. \tag{13}$$

Specifically, the normalized distance metrics are formulated as follows: $\widetilde{c_{p_{ij}}} = (c_{p_{ij}}/c_{p\max})$, $\widetilde{c_{\mathrm{se}_{ij}}} = (c_{\mathrm{se}_{ij}}/c_{\mathrm{se}\max})$, $\widetilde{c_{\mathrm{sa}_{ij}}} = (c_{\mathrm{sa}_{ij}}/c_{\mathrm{sa}\max})$, $c_{p\max}$, $c_{\mathrm{se}\max}$, and $c_{\mathrm{sa}\max}$ denote the maximum spatial distance, maximum spectral Euclidean distance, and maximum spectral angular distance, respectively.

Specifically, the expression (13) maintains the consistency between the pixels and the spectra. By reweighting the spectral distance using spatial edge information, we can further exploit spectral information, thereby successfully avoiding incorrect segmentation caused by neighborhood similarity in the spatial domain.

Algorithm 1 Procedure for Pursuing the Superpixels

Step 1: Input HSI cube Y, set the number of superpixels s, initialize the cube Y with the traditional SLIC method to obtain the initial seed points, set the maximum number of iterations.
Step 2: Calculate the total gradient G with Eq.(12).
Step 3: Repeat
  while *stopping condition is not met* **do**
    Update c_{ij} with Eq.(13).
    Update the seed points with the SLIC method.
  end
Step 4: Output the superpixels $Y^{(1)}$, $Y^{(2)}$, $\cdots$ and $Y^{(s)}$.

For clarity, we show the overall method for tracking the superpixels, taking into account how the interclass pixels are separated at the edge in Algorithm 1. Note that the number of superpixels s is set empirically to avoid the phenomena of over- or under-segmentation. Also, if the seed points do not change repeatedly for ten times, the iterations for Algorithm 1 will stop.

Based on Algorithm 1, the HSI is sensibly segmented into multiple superpixels and unfolded into a series of 2-D HSI matrices. Then, the local graph structure is exploited by addressing each superpixel to overcome the disadvantage of the global manifold structure. Theoretically, the first-order graph and our proposed improved segmentation method can guarantee the homogeneity of the pixels in each superpixel. However, noise and spectral variability can lead to inaccuracies in the first-order graph. Since the first-order neighbor pixels are insufficient to account for the pixels with shared neighborhood structure, we exploit the homogeneity of the pixels in each superpixel to fully exploit the higher-order neighbor relationship, and each pixel in a superpixel is considered as a vertex. In this way, the first-order spatial weight matrix for each superpixel can be computed by the traditional heat kernel. Afterward, the second-order weight is expressed to overcome the shortcomings of the first-order graph, similar to the literature [13], the second-order weight is calculated by computing the correlation of the first-order weight. In other words, the second-order weight represents pixels with a shared neighborhood structure via the first-order graph of shared pixels. Specifically, we create an independent graph for each superpixel, treating each pixel as a vertex. Adjacent pixels are connected by edges whose weights are calculated using the conventional heat kernel function. Additionally, non-adjacent pixels with shared neighborhood configurations are connected by edges whose weights are determined by second-order weights. To sum up, the first-order graph specializes in constructing the relationship of the pixels within the nearest neighborhood; however, the pixels with shared neighborhood structure outside the nearest neighborhood are not represented. On the contrary, the second-order graph specializes in constructing the relationship of the pixels with shared neighborhood structure but outside the nearest neighborhood; thus, the second-order graph remedies the deficiencies of the first-order graph at the cost of weakening the first-order graph. In other words, the second-order graph introduces a new

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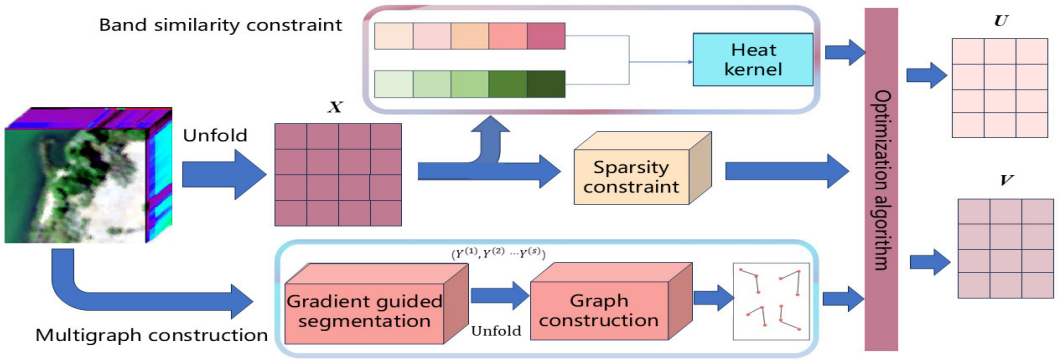


Fig. 3. Flowchart of the proposed method. Specifically, the gradients of the spatial subspaces are derived to constrain the spectral subspace, and the spectral and spatial subspaces are deeply coupled to separate the interclass pixels at the boundary.

problem of how to balance the relationship between the pixels with shared neighborhood structure but outside the nearest neighborhood and the pixels inside the nearest neighborhood.

Finally, both the advantages and disadvantages of the first- and second-order weight matrices are comprehensively considered, and for each superpixel, the higher-order Laplacian matrix is written similarly to the literature [13]. Specifically, let L_h^t be the Laplacian matrix for the tth ($t \in \{1, 2, \ldots, s\}$) superpixel, L_h^t is equal to the linear combination of the first- and the second-order Laplacian matrices. See also the literature [13], L_h^t is written as $\xi\Xi_1^t + (1-\xi)\Xi_2^t$, where Ξ_1^t and Ξ_2^t denote the first-order and second-order Laplacian matrices for the tth superpixel, respectively, which are derived from the first-order and second-order weights of the tth superpixel. ξ is a fine-tuning parameter to control the proportion of the first-order and the second-order Laplacian matrices. Obviously, if ξ is set to a larger value, the first-order Laplacian matrix will contribute more than the second-order Laplacian matrix, and the shortcomings of the first-order graph will not be remedied. On the contrary, a smaller value of ξ can reduce the contribution of the first-order graph but improve the contribution of the second-order graph. In this article, we set $\xi = 0.3$ empirically by referring to the literature [13]. The abundance in each superpixel is optimized using the following higher-order graph constraints, $(1/2)\sum_{t=1}^{s} \mathrm{Tr}(V^t L_h^t (V^t)^{\mathrm{T}})$. Here, s is the number of superpixels and V^t denotes the abundance in the tth superpixel.

C. Proposed Model

Based on the analysis of spectral similarity and the spatial separability, an elaborate weighted factorization model is carried out to guarantee the fidelity of both spectral and spatial subspaces, and to avoid the interferences while capturing the tiny features to a large extent. For clarity, a flowchart for the overall proposed model is shown in Fig. 3. Specifically, to classify the adjacent pixels at the boundary into the correct class and to prevent the local homogeneity from the interferences, the HSI cube is segmented directly with Algorithm 1, and all the obtained subcubes are then unfolded to facilitate the construction of the higher-order graph. Without loss of generality, a common sparsity constraint is used to balance the sparsity and the smoothness of the model. The proposed model is formulated as follows:

$$\begin{aligned}\min_{\Lambda\geq 0, U\geq 0, V\geq 0} & F(\Lambda.U.V) \\ &= \frac{1}{2}\|\Lambda(X - \mathrm{UV})\|_F^2 + \mu\|V\|_{\frac{1}{2}} \\ &\quad + \frac{\lambda}{2}\sum_{t=1}^{s}\mathrm{Tr}\left(V^t L_h^t (V^t)^{\mathrm{T}}\right) + \frac{\rho}{2}\mathrm{Tr}\left(U^{\mathrm{T}}\mathrm{LU}\right) \end{aligned} \tag{14}$$

where μ, λ, and ρ are tradeoff parameters, the abundances are constrained to satisfy the sum-to-one condition. Λ is similar to (5), also an adaptive weight matrix. $\|V\|_{(1/2)}$ is a common sparsity constraint and $\sum_{t=1}^{s}\mathrm{Tr}(V^t L_h^t (V^t)^{\mathrm{T}})$ and $\mathrm{Tr}(U^{\mathrm{T}}\mathrm{LU})$ are the superpixel-guided higher-order graph constraint and the spectral similarity constraint, respectively. For simplicity, the proposed model will be referred to as multigraph regularized nonnegative matrix weighted factorization (MGNMWF) in Section IV–VI.

IV. Optimization

A. Updating Rules

Since the proposed model (14) contains several variables, the variables are optimized alternately. Specifically, given $\Im \geq 0$, $\Re \geq 0$, and $\Omega \geq 0$, suppose them to be the Lagrange multipliers for $\Im \geq 0$, $\Re \geq 0$, and $\Omega \geq 0$, respectively. Then the Lagrange function Γ of (14) can be written as follows:

$$\begin{aligned}\Gamma = & \frac{1}{2}\|\Lambda(X - \mathrm{UV})\|_F^2 + \mu\|V\|_{\frac{1}{2}} + \frac{\lambda}{2}\sum_{t=1}^{s}\mathrm{Tr}\left(V^t L_h^t (V^t)^{\mathrm{T}}\right) \\ & + \frac{\rho}{2}\mathrm{Tr}\left(U^{\mathrm{T}}\mathrm{LU}\right) + \mathrm{Tr}\left(\Im(\Lambda)^{\mathrm{T}}\right) + \mathrm{Tr}\left(\Re(U)^{\mathrm{T}}\right) \\ & + \mathrm{Tr}\left(\Omega(V)^{\mathrm{T}}\right). \end{aligned} \tag{15}$$

According to the above equation, each variable is optimized by fixing the other variables as constants. Specifically, to update U, V, and Λ are considered as the constants. According to the Karush–Kuhn–Tucker (KKT) rule, we can obtain the following equation:

$$\left(-\Lambda^{\mathrm{T}}\Lambda\mathrm{X}\mathrm{V}^{\mathrm{T}} + \Lambda^{\mathrm{T}}\Lambda\mathrm{U}\mathrm{V}\mathrm{V}^{\mathrm{T}} + \rho\mathrm{LU}\right) .* U = 0. \tag{16}$$

Hence, U is optimized by the following rule:

$$U \leftarrow U .* \left(\Lambda^{\mathrm{T}}\Lambda\mathrm{X}\mathrm{V}^{\mathrm{T}} + \rho\mathrm{W}\mathrm{U}\right) ./ \left(\Lambda^{\mathrm{T}}\Lambda\mathrm{U}\mathrm{V}\mathrm{V}^{\mathrm{T}} + \rho\mathrm{D}\mathrm{U}\right) \tag{17}$$

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where $\cdot*$ and $\cdot/$ denote dot product and dot division, respectively.

To optimize V, U, and Λ are fixed, (15) and the rule of KKT can guarantee the following expression, $\left(-U^{\mathrm{T}}\Lambda^{\mathrm{T}}\Lambda X + U^{\mathrm{T}}\Lambda^{\mathrm{T}}\Lambda UV + (\mu/2)V^{-(1/2)} + (\lambda/2)\sum_{t=1}^{s}(\partial \mathrm{Tr}(V^t L_h^t (V^t)^{\mathrm{T}})/\partial V)\right) .* V = 0$. Let f denote the superpixel-guided higher-order graph constraint $\mathrm{Tr}(V^t L_h^t (V^t)^{\mathrm{T}})$. The vec$(\cdot)$ function transforms an arbitrary matrix into a vector. Using the chain rule of differentiation, we derive the following:

$$\frac{\partial f}{\partial \mathrm{vec}\,(V)} = \left(\frac{\partial \mathrm{vec}\,(V^t)}{\partial \mathrm{vec}\,(V)}\right)^{\mathrm{T}} \frac{\partial f}{\partial \mathrm{vec}\,(V^t)}. \tag{18}$$

Since $(\partial f/\partial V^t) = 2V^t L_h^t$, its vectorized form is written as follows:

$$\frac{\partial f}{\partial \mathrm{vec}\,(V^t)} = \mathrm{vec}\left(2V^t L_h^t\right). \tag{19}$$

We construct the selection matrix $\Psi^t \in \mathbb{R}^{P|n^t| \times \mathrm{pn}}$ to extract the abundance vector vec(V^t) of the tth superpixel from the abundance vector vec(V), where $|n^t|$ denotes the number of pixels in the tth superpixel. Specifically, the matrix Ψ^t is composed of $N|n^t|$ block matrices $\Phi \in \mathbb{R}^{P\times P}$, which are defined as follows:

$$\Psi^t = \begin{bmatrix} \Phi_{11} & \cdots & \Phi_{1N} \\ \vdots & \ddots & \vdots \\ \Phi_{|n^t|1} & \cdots & \Phi_{|n^t|N} \end{bmatrix}. \tag{20}$$

Let $\{t_i | 1 \le i \le |n^t|, 1 \le t \le s.\}$ represent the index set of the tth superpixel. Here, t_i represents the column index of the ith column of matrix V^t in matrix V. Φ_{it_i} is set as the size of a $P \times P$ identity matrix, the remaining positions are zero matrix blocks. According to vec$(V^t) = \Psi^t \mathrm{vec}(V)$, (18) can be further derived as follows:

$$\frac{\partial f}{\partial \mathrm{vec}\,(V)} = \left(\Psi^t\right)^{\mathrm{T}} \mathrm{vec}\left(2V^t L_h^t\right). \tag{21}$$

Finally, the partial derivative of the superpixel-guided higher-order graph constraint f with respect to the abundance matrix V is derived as follows:

$$\begin{aligned} \frac{\partial f}{\partial V} &= \mathrm{vec}^{-1}\left(\left(\Psi^t\right)^{\mathrm{T}} \mathrm{vec}\left(2V^t L_h^t\right)\right) \\ &= \mathrm{vec}^{-1}\left(\left(\Psi^t\right)^{\mathrm{T}} \mathrm{vec}\left(2V^t D_h^t\right)\right) \\ &\quad - \mathrm{vec}^{-1}\left(\left(\Psi^t\right)^{\mathrm{T}} \mathrm{vec}\left(2V^t W_h^t\right)\right). \end{aligned} \tag{22}$$

Thus, V is optimized according to the following rule:

$$\begin{aligned} V \leftarrow V .* \Bigg(& U^{\mathrm{T}}\Lambda^{\mathrm{T}}\Lambda X \\ & + \lambda \sum_{t=1}^{s} \mathrm{vec}^{-1}\left(\left(\Psi^t\right)^{\mathrm{T}} \mathrm{vec}\left(V^t W_h^t\right)\right)\Bigg) ./ \Bigg(U^{\mathrm{T}}\Lambda^{\mathrm{T}}\Lambda UV \\ & + \frac{\mu}{2} V^{-\frac{1}{2}} + \lambda \sum_{t=1}^{s} \mathrm{vec}^{-1}\left(\left(\Psi^t\right)^{\mathrm{T}} \mathrm{vec}\left(V^t D_h^t\right)\right)\Bigg) \end{aligned} \tag{23}$$

where D_h^t is a diagonal matrix and is given by $D_{h_{ii}}^t = \sum_{j=1}^{|n^t|} W_{h_{ij}}^t$, and $i, j \in \{1, 2, \ldots, |n^t|\}.$ Similarly, to update Λ, U and V are fixed, and according to the KKT, we can obtain the following expression $(\Lambda XX^{\mathrm{T}} - \Lambda UVX^{\mathrm{T}} - \Lambda XV^{\mathrm{T}}U^{\mathrm{T}} + \Lambda UVV^{\mathrm{T}}U^{\mathrm{T}}) .* \Lambda = 0$, then Λ is updated as follows:

$$\begin{aligned} \Lambda \leftarrow \Lambda .* & \left(\Lambda UVX^{\mathrm{T}} + \Lambda XV^{\mathrm{T}}U^{\mathrm{T}}\right) \\ & ./ \left(\Lambda XX^{\mathrm{T}} + \Lambda UVV^{\mathrm{T}}U^{\mathrm{T}}\right). \end{aligned} \tag{24}$$

Algorithm 2 Overall Procedure for the Proposed Model

Step 1: Input HSI matrix $X \in \mathbb{R}^{b\times N}$, the superpixels $Y^{(1)}, Y^{(2)}, \cdots, Y^{(s)}$, the number of endmembers P, the trade-off parameters μ, λ, ρ, and the maximum number of iterations $MIter$.

Step 2: Initialize Λ as an identity matrix. Initialize endmember and abundance matrices by VCA [25] and FCLS [14] respectively. Calculate the weight matrix W by Eq.(6), unfold each superpixel and calculate the weight matrix of each superpixel by the way of Eq.(6).

Step 3: Repeat
while *stopping condition is not met* **do**
Update U with Eq.(17).
Update V with Eq.(23).
Update Λ with Eq.(24)
end

Step 4: Output endmember matrix $U \in \mathbb{R}^{b\times P}$ and abundance matrix $V \in \mathbb{R}^{P\times N}$.

For clarity, the detailed optimization rules are shown in Algorithm 2. Specifically, the vertex component analysis (VCA) [25] and the fully constrained least-squares (FCLS) [14] are the canonical methods for computing the endmember matrix and abundance matrix, respectively. To ensure the convergence of the Algorithm 2, the maximum number of iterations is set to 1000, and an error tolerance is set as $1e^{-4}$, that is, if the error of the proposed model 14 is no longer than the set tolerance successively for ten times, the procedure stops. Otherwise, the updating process is repeated until the number of 1000 iterations are reached.

B. Convergence Analysis

The proof of convergence of the algorithm we proposed in Algorithm 2 is as follows.

Theorem 1: Algorithm 2 monotonically decreases the objective function value of problem (14) at each iteration until convergence.

Proof: The Λ, U, and V variables in the $k+1$ iteration are represented by Λ^{k+1}, U^{k+1}, and V^{k+1}, respectively. Since updating Λ is equivalent to minimizing the objective function $\min_{\Lambda\ge 0}(1/2)\|\Lambda(X - \mathrm{UV})\|_F^2$, we can obtain

$$\frac{1}{2}\|\Lambda^{k+1}\left(X - U^{k+1}V^{k+1}\right)\|_F^2 \le \frac{1}{2}\|\Lambda^{k}\left(X - U^{k+1}V^{k+1}\right)\|_F^2. \tag{25}$$

Let $F_1(\Lambda, U, V) = (1/2)\|\Lambda(X - \mathrm{UV})\|_F^2$ and $F_2(U, V) = \mu\|V\|_{(1/2)} + (\lambda/2)\sum_{t=1}^{s}\mathrm{Tr}(V^t L_h^t (V^t)^{\mathrm{T}}) + (\rho/2)\mathrm{Tr}(U^{\mathrm{T}}\mathrm{LU})$, then $F(\Lambda, U, V) = F_1(\Lambda, U, V) + F_2(U, V)$. We can obtain the update for$U$

$$\begin{aligned} & F_1\left(\Lambda^k, U^{k+1}, V^k\right) + F_2\left(U^{k+1}, V^k\right) \\ & \quad \le F_1\left(\Lambda^k, U^k, V^k\right) + F_2\left(U^k, V^k\right). \end{aligned} \tag{26}$$

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TABLE I
CHARACTERISTICS OF THE METHODS COMPARED

$L_{1/2}$-NMF	A canonical blind HU method, which used $L_{1/2}$ to pursue the sparse unmixing results [10].
gtvMBO	A more recent method than $L_{1/2} - NMF$, which is unlike $L_{1/2} - NMF$, as gtvMBO used the graph total variation to constrain the abundance [21].
SDNMF-TV	Unlike the traditional NMF-based unmixing methods, SDNMF-TV used the deep NMF as the HU framework, and to pursue the sparsity, total variation was employed to constrain the deep NMF framework [11].
SSAINMF	SSAINMF was published in 2023 year, both spectral and spatial information were exploited to construct an anti-interference NMF for the task of HU [12].
HGNMF-FS	HGNMF-FS was also published in 2023 year. To select the feature adaptively, HGNMF-FS used higher-order graph to constrain the NMF [13].
MAC-U	It used deep autoencoder networks for the task of HU with the consideration of the nonliearity [16].
GL1-DNMF	GL1-DNMF implements noise filter guidance constraints and very common constraints respectively on the shallow layer and the gradually deepening layers based on deep NMF [19].

Similarly, updating V can be achieved

$$F_1\left(\Lambda^k, U^{k+1}, V^{k+1}\right) + F_2\left(U^{k+1}, V^{k+1}\right) \leq F_1\left(\Lambda^k, U^{k+1}, V^k\right) + F_2\left(U^{k+1}, V^k\right). \tag{27}$$

We can obtain the following by combining the two equations above:

$$F_2\left(U^{k+1}, V^{k+1}\right) \leq F_1\left(\Lambda^k, U^k, V^k\right) + F_2\left(U^k, V^k\right) - F_1\left(\Lambda^k, U^{k+1}, V^{k+1}\right). \tag{28}$$

Summing (25) and (28) on both sides, and finally obtain

$$F\left(\Lambda^{k+1}, U^{k+1}, V^{k+1}\right) \leq F\left(\Lambda^k, U^k, V^k\right). \tag{29}$$

Since the loss value of problem (14) is greater than 0, the optimization algorithm is convergent. □

V. EXPERIMENTS

To validate the proposed model and algorithm, several recent and canonical methods are compared on both synthetic and real datasets. Specifically, the methods compared are $L_{1/2}$–NMF [10], gtvMBO [21], SDNMF–TV [11], SSAINMF [12], HGNMF–FS [13], MAC–U [16], and GL1–DNMF [19]. The characteristics of the methods compared are listed in Table I. In other words, except for the method MAC–U, which used the deep networks, the other compared methods used traditional NMF on deep NMF as the HU framework with different constraints. It is worth noting that all the involved parameters of the compared methods are set according to their original literature. Since the problem of HU is ill-posed, we ran each method 15 times repeatedly. The means and deviations of the metric values are recorded, and we mark the best results in bold. Furthermore, for fairness, each comparison is performed using a uniform initializing strategy.

To quantitatively evaluate the performance of the proposed method, two broadly used metrics, SAD and root-mean-square error (RMSE), are employed. Specifically, the metrics can be defined as follows:

$$\text{SAD} = \sum_{i=1}^{P} \arccos\left(\frac{u_i^{\mathrm{T}} \tilde{u}_i}{\|u_i^{\mathrm{T}}\| \cdot \|\tilde{u}_i\|}\right) \tag{30}$$

$$\text{RMSE} = \left(\frac{1}{N}\sum_{i=1}^{N} \|v_i - \tilde{v}_i\|_2^2\right)^{\frac{1}{2}}. \tag{31}$$

where u_i and $\tilde{u}_i$ are the ith true and estimated endmember vectors, respectively, and v_i and $\tilde{v}_i$ are the ith true and estimated abundance vectors, respectively.

A. On Synthetic Datasets

In this section, two synthetic datasets are generated and named SD1 and SD2, respectively. The detailed information of the SD1 and SD2 is illustrated as follows.

1) *SD1:* Five spectra are randomly selected from the United States Geological Survey (USGS) spectral library [26] as the endmembers. The abundance maps are then generated using the selected five spectra. Specifically, the abundance maps are generated according to the distribution of the real HSI, and the conditions of the nonnegativity and sum-to-one are guaranteed. Finally, the synthetic HSI data are obtained by multiplying the selected spectra and the generated abundances. To approximate the HSI in the real scene, the additive noise is added to the products of the selected spectra and the generated abundances. The final synthetic data consists of 224 bands, and the spatial size is 75×75.
2) *SD2:* For the second synthetic dataset SD2, 5 endmembers are also randomly selected from the USGS library. Unlike the SD1, we employ a Gaussian function to generate the abundances and guarantee the nonnegativity and sum-to-one conditions. To validate the anti-interference of the proposed model, the pervasive interferences are simulated. Specifically, both the endmember variability and the additive noise are simulated according to the work in the literature [27]. To simulate the endmember variability, the scale factors are in the range of $[0.75, 1.25]$. Finally, the simulated dataset SD2 has 224 channel spectra and the spatial size is 200×200.

1) Parameter Analysis: In this section, the parameters involved, s, μ, λ, and ρ are studied using the SD1 is used. Specifically, to study the parameters listed above, the SD1 is simulated with 30 dB of additive noise.

To study the influence of the number of superpixels, μ, λ, and ρ are fixed at $1e^{-5}$, 0.001, and 0.001, respectively. The number of superpixels s is set in the range of $[70, 120]$. The results are shown in Fig. 4. The results in Fig. 4 show that both metrics of SAD and RMSE can achieve the best values at the superpixel number of 100. In theory, a

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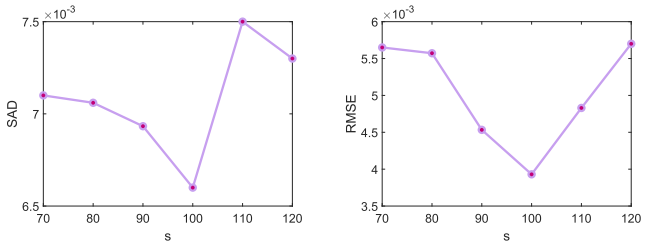


Fig. 4. (a) Estimated SAD values on the SD1 with a varying number of superpixels. (b) Estimated RMSE values on the SD1 with a varying number of superpixels.

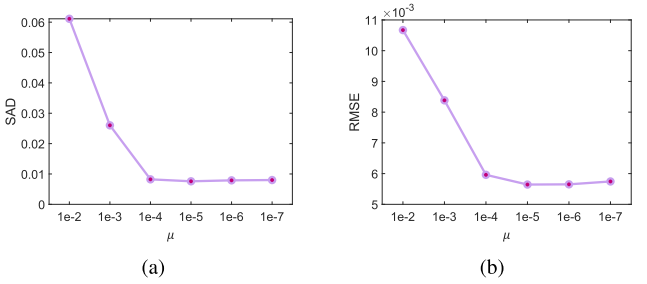


Fig. 5. Estimated results on the SD1 with varied parameter μ. (a) Estimated SAD values on the SD1 with a varying parameter μ. (b) Estimated RMSE values on the SD1 with a varying parameter μ.

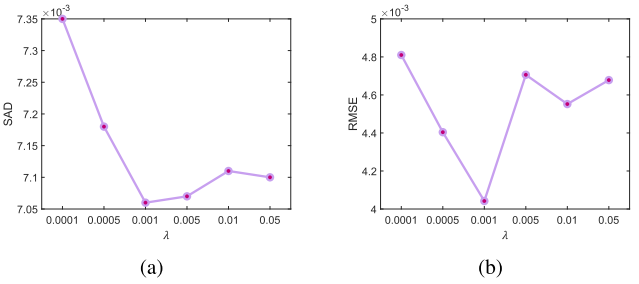


Fig. 6. Estimated results on the SD1 with varied parameter λ. (a) Estimated SAD values on the SD1 with a varying parameter λ. (b) Estimated RMSE values on the SD1 with a varied λ parameter.

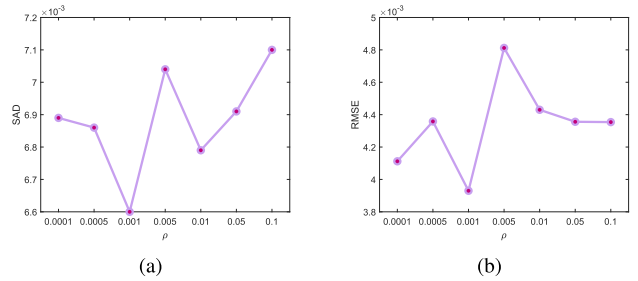


Fig. 7. Estimated results on the SD1 with varied parameter ρ. (a) Estimated SAD values on the SD1 with a varied parameter ρ. (b) Estimated RMSE values on the SD1 with a varied parameter ρ.

larger value of s means that the HSI cube is divided into more smaller subcubes; as a result, the numerous original homogeneous regions are oversegmented into different fine regions, and these fine regions are difficult to recover to the raw HSI cube as multiple redundant edge information is introduced by the oversegmentation. In other words, the global information is not captured as the large-scale information is unduly separated with additive edges. Therefore, these over-segmented subcubes cause the unmixing results to be worse. On the contrary, a smaller value of s causes the phenomenon of under-segmentation, that is, each obtained subcube may contain heterogeneous information, then the heterogeneity will misguide the subsequent iterations for unmixing, as the under-segmentation usually neglects the latent subtle cues in the original HSI cube. Moreover, the under-segmentation leads to the situation of interclass inseparability, which plays a negative role in the task of HU.

Next, the other three parameters are studied, with s fixed at 100. To study the tradeoff parameter μ, except for the number of superpixels s, both the parameters λ and ρ are fixed at 0, and the parameter μ is set in the range of $[1e^{-2}, 1e^{-7}]$. The experimental results are shown in Fig. 5. It is obvious that the results of the SAD and RMSE become better with the increasing values of the parameter μ. However, when the parameter μ is greater than $1e^{-5}$, the values of SAD and RMSE become worse than the results when $\mu = 1e^{-5}$.

Similarly, the remaining parameters λ and ρ are studied in an ordered fashion, that is, to study λ, μ, and ρ are set to $1e^{-5}$ and 0, respectively, λ is set in the range of $[0.0001, 0.05]$. For ρ, μ, and λ are set as $1e^{-5}$ and 0.0001, respectively, and ρ is set in the range of $[0.0001, 0.1]$. The results with varied parameters λ and ρ are, respectively, shown in Figs. 6 and 7. According to Figs. 6 and 7, when λ is set as 0.001 and ρ is also set as 0.001, the results of the SAD and RMSE are better than the other values of λ and ρ.

Theoretically, the parameters μ, λ, and ρ influence the sparsity, spectral smoothness, and spatial smoothness of the unmixing results, respectively. Specifically, a larger value of μ will lead to over-sparsity and will suppress some effective information from being expressed, inevitably missing some important features. Conversely, a smaller value of μ will make the unmixing model sensitive to the noise. Regarding the parameters λ and ρ, these two parameters balance the smoothness from the spectral and spatial domains, respectively. Since the reflectance values of the adjacent bands are very similar, it is necessary to maintain the spectral smoothness, so the weight Λ is restrained to prevent the distortion. Similarly, the parameter ρ aims to control the spatial smoothness under the guidance of the multiple higher-order graphs. Therefore, larger values of λ and ρ will cause over-smoothness and miss the fine discriminative cues; conversely, smaller values of λ and ρ will cause the distortion, which is inconsistent with the real scene.

Overall, both the segmentation and the local higher-order constraint play a role in achieving the separability of the interclass pixels at the boundary, that is, a larger value of the parameter λ often requires a smaller value of the parameter s in order for the proposed model to run coordinately. Conversely, a larger value of the parameter s requires a smaller value of the parameter λ. Furthermore, the proposed model MGNMWF is based on a weighted factorization to select the reconstruction feature UV by an adaptive update parameter Λ. Thus, a larger value of the parameter λ often requires a larger value of the parameter ρ to balance the selection of the reconstruction feature UV; otherwise, a smaller value of the parameter λ requires a smaller value of the parameter ρ.

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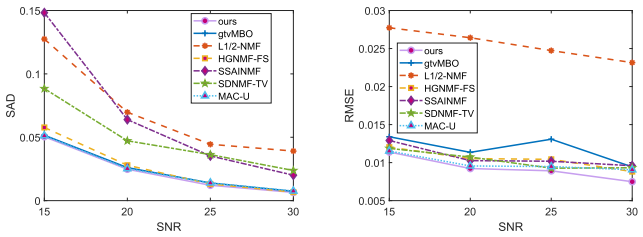


Fig. 8. Metric values of different methods with the varied noises. (a) SAD values with the varied noises. (b) RMSE values with the varied noises.

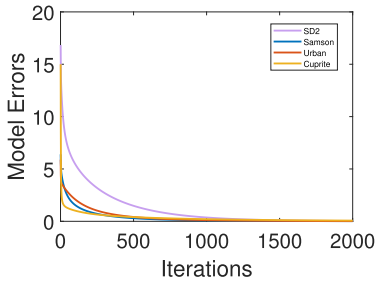


Fig. 9. Model errors with the different number of iterations on the different datasets.

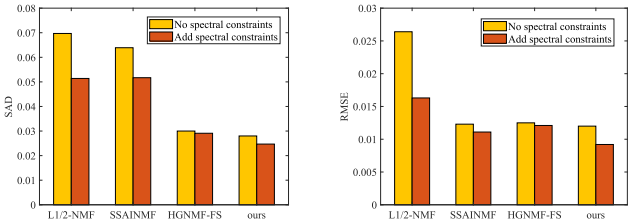


Fig. 10. Comparison of (a) SAD and (b) RMSE of different algorithms after adding a spectral similarity regularizer.

Based on the above analysis, the parameters s, μ, λ, and ρ were set as 100, $1e^{-5}$, 0.001, and 0.001, respectively, in all the subsequent experiments on the synthetic datasets.

2) Robustness Analysis: To validate the robustness of the proposed model, we add different proportions of noise to the dataset SD1, that is, the values of the signal-to-noise ratio (SNR) are 15, 20, 25, and 30 dB, respectively. The results are shown in Fig. 8. It is obvious that almost all the methods perform better with increasing values of the SNR. In particular, the proposed model MGNMWF can always outperform the other methods. The results in Fig. 8 show that the proposed model is more robust than the other methods. Specifically, the SAD values of the MAC-U method are almost equal to our proposed model, but it performs worse in estimating abundances with the decreasing of the noise decreases. The SAD value of SDNMF-TV is much higher than that of our algorithm under different noise conditions. Additionally, our proposed method MGNMWF is also superior to the two recently published methods, SSAINMF and HGNMF–FS. According to Table I, SDNMF-TV introduces TV regularization to deep NMF. SSAINMF is an anti-interference method, and HGNMF–FS employs a higher-order graph to guide the weighted NMF. Unlike these three methods, our proposed method MGNMWF considers the spectral similarity to prevent the over-sparsity through the weighted factorization. Additionally, the proposed method, MGNMWF, fully exploits the relationship between the adjacent pixels within a neighborhood and achieves the goal of interclass separability by a spectral–spatial coupling method. In conclusion, the weighted factorization helps select the features, but at the cost of degrading the smoothness of the spectra. The higher-order graph regularizer can maintain the spatial smoothness, but neglects the interclass separability of the pixels within a neighborhood.

3) Convergence Analysis: To study the convergence of the model, we plot the model error curves on both the synthetic and real datasets with increasing iteration times. As shown in Fig. 9, the curves clearly demonstrate that model errors tend to stabilize when the number of iterations is greater than or equal to 1000. Thus, the model errors decrease monotonically with increasing iteration times, and it can converge well, so we set the maximum number of iterations to 1000 in all the experiments.

4) Spectral Similarity Analysis: Experiments were conducted on dataset SD1 at an SNR of 20 dB to validate the effectiveness of the spectral similarity constraint. The spectral similarity regularization term was incorporated into several NMF-based unmixing methods. Fig. 10 compares the performance of different algorithms with and without the spectral similarity regularizer. As can be seen, adding the spectral similarity regularizer reduced both the SAD and RMSE values across all algorithms. This validates the beneficial effect of the spectral similarity regularizer for HU. The regularizer ensures spectral smoothness, enhancing the algorithm's ability to leverage the complete spectral information in the spectral domain.

5) Ablation Study: To perform in this section, the dataset SD1 is used, and the value of SNR is set as 20 dB. The baseline model, the baseline model with adaptive weights, the baseline model with spectral constraint, and the baseline model with multiple spatial constraints are tested, respectively. Specifically, four different models under test are defined as follows:

$$\min_{\Lambda\geq 0,U\geq 0,V\geq 0} \frac{1}{2}\|(X-\mathrm{UV})\|_F^2+\mu\|V\|_{\frac{1}{2}} \tag{32}$$

$$\min_{\Lambda\geq 0,U\geq 0,V\geq 0} \frac{1}{2}\|\Lambda(X-\mathrm{UV})\|_F^2+\mu\|V\|_{\frac{1}{2}} \tag{33}$$

$$\min_{\Lambda\geq 0,U\geq 0,V\geq 0} \frac{1}{2}\|(X-\mathrm{UV})\|_F^2+\mu\|V\|_{\frac{1}{2}} + \frac{\lambda}{2}\sum_{t=1}^{s}\mathrm{Tr}\left(V^t L_h^{(t)}\left(V^t\right)^{\mathrm{T}}\right) \tag{34}$$

$$\min_{\Lambda\geq 0,U\geq 0,V\geq 0} \frac{1}{2}\|(X-\mathrm{UV})\|_F^2+\mu\|V\|_{\frac{1}{2}} + \frac{\rho}{2}\mathrm{Tr}\left(U^{\mathrm{T}}\mathrm{LU}\right). \tag{35}$$

To further test the influence of multiple higher-order graph constraints, the following model incorporates these con-

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TABLE II
RESULTS OF THE ABLATION STUDY

	Model (32)	Model (33)	Model (34)	Model (35)	Model (36)	Model (37)	Overall Model (14)
SAD	0.0337	0.0295	0.0284	0.0312	0.02648	0.0265	**0.02478**
RMSE	0.01482	0.01283	0.01427	0.0132	0.01239	0.01077	**0.00922**

TABLE III
RESULTS OF DIFFERENT METHODS ON DATASET SD2

	$L_{1/2}-NMF$	$gtvMBO$	$SDNMF-TV$	$SSAINMF$	$HGNMF-FS$	$MAC-U$	$GL1-DNMF$	$ours$
SAD	0.0383 ± 0.0022	0.0168 ± 0.0002	0.0807 ± 0.0135	0.0705 ± 0.006	0.0154 ± 0.0023	0.0114 ± 0.001	0.0153 ± 0.0024	**0.0088±0.0002**
RMSE	0.0016 ± 0.0003	0.0016 ± 0.0003	0.0031 ± 0.0021	0.0041 ± 0.0021	0.0023 ± 0.0001	0.0015 ± 0.0004	0.0015 ± 0.0004	**0.0008±0.0001**
Time(s)	0.7414	3.0971	311.2447	**0.6413**	123.887	232.6182	79.3419	86.9901

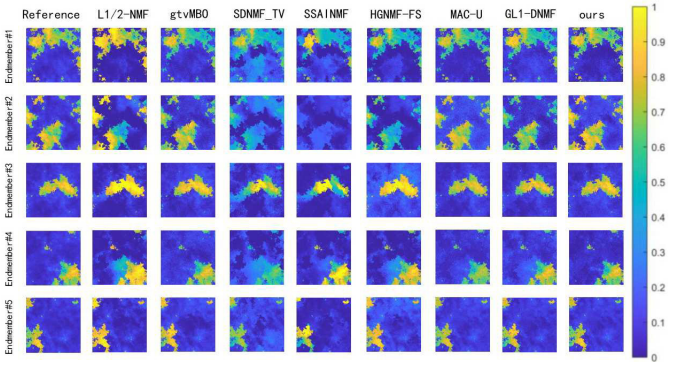


Fig. 11. Estimated abundance maps for dataset SD2 by different methods.[1]

straints into the baseline model with adaptive weights. The mathematical expression is written as follows:

$$\min_{\Lambda\geq 0,U\geq 0,V\geq 0} \frac{1}{2}\|\Lambda(X-\mathrm{UV})\|_F^2+\mu\|V\|_{\frac{1}{2}} + \frac{\lambda}{2}\sum_{t=1}^{s}\mathrm{Tr}\left(V^t L_h^{(t)}\left(V^t\right)^{\mathrm{T}}\right). \tag{36}$$

Similarly, the spectral similarity constraint is tested by adding the spectral smoothness constraint to the baseline model with adaptive weights, and the model is formulated mathematically as follows:

$$\min_{\Lambda\geq 0,U\geq 0,V\geq 0} \frac{1}{2}\|\Lambda(X-\mathrm{UV})\|_F^2+\mu\|V\|_{\frac{1}{2}} + \frac{\rho}{2}\mathrm{Tr}\left(U^{\mathrm{T}}\mathrm{LU}\right). \tag{37}$$

The results of the ablation experiments are shown in Table II. The baseline model (32) performed worse than the other models, demonstrating the significance of adaptive weighting mechanisms, the multiple spatial graph constraints, and the spectral constraint to the task of HU. Models (36) and (37) achieve significant performance improvements by incorporating multiorder graph constraints and spectral similarity constraints, respectively, into baseline mode (33), which is equipped with adaptive weighting mechanisms. Furthermore, models (34) and (35), which lack adaptive weighting mechanisms, exhibit significantly inferior performance compared to their counterparts (36) and (37). Specifically, the designed spatial and spectral constraints contribute equally to the extraction of the spectra. When the baseline model with adaptive weights is equipped with both the spatial and spectral constraints, the metric results of the SAD and RMSE are better than those of the other models. In short, the baseline model fails to incorporate feature selection mechanisms and neglects critical issues such as spectral smoothness and local spatial coherence; the overall proposed model (14) works harmoniously by driving the feature-selected baseline model with the two proposed constraints.

6) Comparative Experiments: The comparative experiments are performed on the dataset SD2 with the value of SNR 25dB, the results of our proposed model and the compared methods are shown in Table III and Fig. 11, respectively.[1] According to Fig. 11, the recovered abundance maps of our proposed model MGNMWF and MAC-U coincide with the reference better than the other methods. However, the method MAC-U recovers the endmembers 1# and 4# worse than our proposed method MGNMWF. The value of the RMSE in Table III also indicates that the proposed method MGNMWF can obtain better abundance maps than the other methods. In terms of endmember extraction, Table III shows that our proposed method MGNMWF is far superior to the other methods compared. Although our proposed method MGNMWF performs better in both endmember extraction and abundance estimation, the time cost of the proposed method MGNMWF is higher than the methods $L_{1/2}$–NMF, gtvMBO, SSAINMF, and GL1-DNMF. The reason is that the proposed method MGNMWF needs to capture the higher-order relationship of the pixels in each superpixel and to calculate the weight matrix of the spectral similarity.

B. On Real Datasets

To further validate the performance of the proposed method, MGNMWF, three widely used real datasets are employed; that is, the proposed method, MGNMWF, and the compared methods are implemented on Samson, Urban, and Cuprite datasets, respectively. Specifically, the parameter μ is set as $1e^{-5}$ for all the following experiments on the real datasets; the other parameters s, λ, and ρ are set differently on the different real datasets.

[1]A bright yellow pixel indicates a significant contribution from the material, while a dark blue pixel indicates no contribution at all. Specifically, pure yellow represents a material contribution of 1 for a given class in that pixel, while pure blue represents a contribution of 0.

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TABLE IV
VALUES OF THE SAD BY DIFFERENT METHODS ON THE SAMSON DATASET

	$L_{1/2}-NMF$	$gtvMBO$	$SDNMF-TV$	$SSAINMF$	$HGNMF-FS$	$MAC-U$	$GL1-DNMF$	$ours$
Soil	0.0222 ± 0.0007	0.034 ± 0.0207	0.0227 ± 0.0327	0.0368 ± 0.0085	0.0246 ± 0.0007	**0.0207±0.0005**	0.0227 ± 0	0.033 ± 0.009
Tree	0.0357 ± 0.0021	0.0704 ± 0.0144	0.0985 ± 0.0391	0.0433 ± 0.0153	0.0501 ± 0.0063	0.0496 ± 0.0001	0.0626 ± 0.0013	**0.0303±0.0002**
Water	0.3071 ± 0.0585	0.3439 ± 0.0402	0.1184 ± 0.0615	0.1641 ± 0.0334	0.1033 ± 0.0223	0.1292 ± 0.0009	**0.0871±0.0012**	0.1019 ± 0.001
Mean	0.1217 ± 0.02	0.1494 ± 0.0113	0.0799 ± 0.0143	0.0814 ± 0.0138	0.0593 ± 0.0073	0.0665 ± 0.0005	0.0575 ± 0.0004	**0.0551±0.0004**

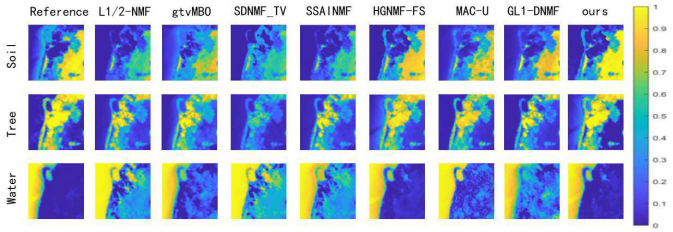


Fig. 12. Estimated abundance maps for the Samson dataset by different methods.

1) Samson: The Samson dataset has 156 bands, the wavelength is in the range of [0.401, 0.889] μm, and the spatial size is 95 $\times$ 95. According to [28], the Samson dataset consists of three endmembers, that is, soil, tree, and water.

Since the mixing degree of the Samson dataset is milder than that of the synthetic datasets SD1 and SD2, we set the number of superpixels in the Samson dataset as 50, while setting larger values of λ, ρ as 0.05. The reason is that the smaller number of superpixels tends to misclassify the neighboring pixels at the boundary, so a larger value of the λ parameter is set, while a larger value of the ρ parameter is set. The results are presented in Table IV. As can be seen, our method outperforms the others in extracting the endmember of the tree. Furthermore, our proposed method, MGNMWF, achieves better average endmember extraction results. In terms of estimation accuracy, the proposed method exhibits superior performance with a lower bias than most algorithms. Although the bias value of GL1-DNMF is equal to that of our method, its SAD metric is substantially higher than that of our approach. This suggests that, despite its refined bias control, it has compromised spectral fidelity in endmember extraction. Additionally, the estimated abundance maps by different methods are shown in Fig. 12. According to the reference abundance maps, it is clear that the proposed method MGNMWF recovers the abundance maps better than the other methods. Specifically, HGNMF–FS recovers maps that are slightly inferior to ours, and the methods $L_{1/2}$–NMF, gtvMBO, SDNMF–TV, SSAINMF, MAC–U, and GL1–DNMF fail to estimate the maps of "soil." On the basis of Table IV and Fig. 12, we can conclude that the proposed method MGNMWF performs better than the other methods compared, both in the extraction of the endmembers and in the estimation of the abundance maps.

2) Urban: The Urban dataset is a common dataset for the HU task. It was collected by a hyperspectral digital imagery collection experiment (HYDICE) sensor [29]. There are 218 channels and the image size is 307 $\times$ 307, the wavelength of the bands is in the range of [0.4–2.5] μm

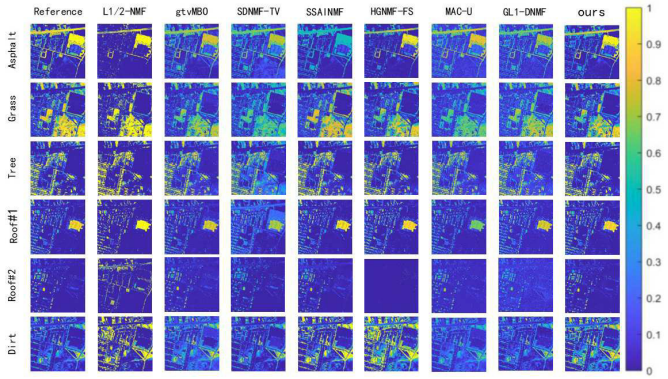


Fig. 13. Estimated abundance maps by different methods on the urban dataset.[1]

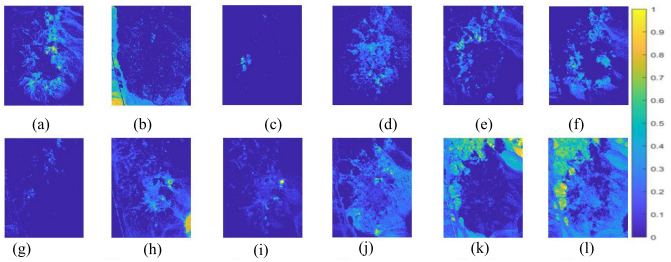


Fig. 14. Estimated abundance maps by different methods for the cuprite dataset.[1] (a) Alunite. (b) Andradite. (c) Buddingtonite. (d) Chalcedony. (e) Dumortierite. (f) Kaolinite-2. (g) Nontronite. (h) Montmorillonite. (i) Muscovite. (j) Kaolinite-1. (k) Pyrope. (l) Sphene.

[30], a subimage of 256 $\times$ 256 is cropped for the following experiments. Generally, this subimage is considered to contain 6 endmembers, that is, asphalt, grass, tree, roof#1, roof#2, and dirt. Additionally, according to the existing [30], we remove the bands contaminated by the water vapor absorption and atmospheric interference, leaving 162 bands for the following experiments.

Since the image size of the Urban dataset is larger than that of the Samson dataset, the number of superpixels is set as $s = 100$. Additionally, because the spatial structure of the Urban dataset is more complex than that of the synthetic datasets SD1 and SD2, the parameter λ on the Urban dataset is set slightly larger than that of the synthetic dataset, and $\lambda = 0.005$. The parameter ρ is still set as 0.001. The quantitative metrics and abundance maps are shown in Table V and Fig. 13, respectively. According to the reference maps, $L_{1/2}$–NMF performed the worst at recovering the maps. The gtvMBO, SDNMF–TV, MAC–U, and GL1–DNMF methods also inaccurately recovered the asphalt and grass maps. Moreover, SDNMF–TV and MAC–U did not recover the map

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TABLE V

VALUES OF THE SAD BY DIFFERENT METHODS ON THE URBAN DATASET

	$L_{1/2}-NMF$	$gtvMBO$	$SDNMF-TV$	$SSAINMF$	$HGNMF-FS$	$MAC-U$	$GL1-DNMF$	$ours$
Asphalt	0.0932 ± 0.0382	0.1712 ± 0.0465	0.2064 ± 0.01	0.1014 ± 0.0266	0.0226 ± 0.0125	0.1294 ± 0.0216	0.1150 ± 0.0121	**0.0142±0.0019**
Grass	0.0966 ± 0.0158	0.1022 ± 0.0455	0.1038 ± 0	0.1086 ± 0.0046	0.0577 ± 0.0386	0.1023 ± 0.034	0.0680 ± 0.0308	**0.0557±0.0331**
Tree	0.0773 ± 0.0235	0.1035 ± 0.0229	0.1023 ± 0.0086	0.0667 ± 0.0139	**0.0335±0.0031**	0.0362 ± 0.0088	0.0982 ± 0.0246	0.0414 ± 0.01
Roof#1	0.0823 ± 0.0315	0.0636 ± 0.0164	0.1285 ± 0.0489	0.0549 ± 0.0395	**0.0406±0.0163**	0.1143 ± 0.0214	0.1005 ± 0.0162	0.0407 ± 0.01
Roof#2	0.1261 ± 0.0534	0.0585 ± 0.012	0.0328 ± 0.0013	0.0944 ± 0.1604	0.1321 ± 0.0707	0.0349 ± 0.019	0.1206 ± 0.0278	**0.0319±0.0027**
Dirt	0.1068 ± 0.0158	0.2190 ± 0.0626	0.0608 ± 0.0505	0.0846 ± 0.0254	0.0639 ± 0.0359	0.1457 ± 0.0255	**0.0312±0.0068**	0.1368 ± 0.04
Mean	0.0971 ± 0.0461	0.1197 ± 0.0071	0.1058 ± 0.0166	0.0851 ± 0.0366	0.0584 ± 0.0205	0.0938 ± 0.0188	0.0889 ± 0.0071	**0.0535±0.0021**

TABLE VI

SAD VALUES OF DIFFERENT METHODS ON THE CUPRITE DATASET

	$L_{1/2}-NMF$	$gtvMBO$	$SDNMF-TV$	$SSAINMF$	$HGNMF-FS$	$MAC-U$	$GL1-DNMF$	$ours$
Alunite	0.0708 ± 0.0156	0.1054 ± 0	0.0933 ± 0.0016	0.1434 ± 0.0093	0.0898 ± 0.002	0.1091 ± 0.0098	0.1055 ± 0.0202	**0.069±0.00013**
Andradite	0.1441 ± 0.0094	0.1269 ± 0.029	0.0847 ± 0.0261	0.4501 ± 0.0173	0.0797 ± 0.0273	0.0634 ± 0.0125	0.0956 ± 0.038	**0.0617±0.0277**
Buddingtonite	0.0743 ± 0.0199	0.0979 ± 0.0308	**0.0678±0.0229**	0.1874 ± 0.0066	0.1283 ± 0.0128	0.0804 ± 0.0123	0.1207 ± 0.0331	0.0809 ± 0.0035
Dumortierite	0.2114 ± 0.0731	0.2480 ± 0.02	0.1305 ± 0.0756	0.1400 ± 0.0007	0.1543 ± 0.0142	0.2024 ± 0.0113	0.1336 ± 0.0901	**0.1183±0.0086**
Kaolinite−1	0.0667 ± 0.0118	0.0677 ± 0.02	0.0803 ± 0.0045	**0.0627±0.0149**	0.0773 ± 0.0127	0.0725 ± 0.0002	0.1399 ± 0.0552	0.1012 ± 0.0054
Kaolinite−2	0.0863 ± 0.0064	0.0712 ± 0.02	**0.0509±0.0143**	0.0996 ± 0.0036	0.0724 ± 0.0037	0.0809 ± 0.001	0.0944 ± 0.1142	0.1380 ± 0.0074
Muscovite	**0.0588±0.0361**	0.0641 ± 0	0.0736 ± 0.0161	0.1353 ± 0.0247	0.0653 ± 0.0234	0.0767 ± 0.0024	0.1100 ± 0.1042	0.0696 ± 0.0081
Montmorillonite	0.1411 ± 0.0097	**0.0565±0**	0.0993 ± 0.037	0.1288 ± 0.0005	0.0931 ± 0.0128	0.1033 ± 0.0054	0.0703 ± 0.0635	0.1023 ± 0.0047
Nontronite	0.1082 ± 0.0161	0.1188 ± 0	0.1203 ± 0.0638	**0.0950±0.0101**	0.1108 ± 0.0139	0.1269 ± 0.0047	0.1147 ± 0.0838	0.0994 ± 0.0156
Pyrope	0.0915 ± 0.0031	0.1287 ± 0	**0.0876±0.0291**	0.2198 ± 0.0029	0.1365 ± 0.0306	0.1170 ± 0.0059	0.1561 ± 0.1337	0.1084 ± 0.0137
Sphene	0.0622 ± 0.0473	**0.0556±0**	0.1087 ± 0.0069	0.2458 ± 0.0251	0.0721 ± 0.0488	0.0688 ± 0.0189	0.1211 ± 0.07	0.1140 ± 0.0052
Chalcedony	0.1116 ± 0.0069	0.1962 ± 0	0.2676 ± 0.0291	0.1741 ± 0.0062	0.1009 ± 0.0086	0.0625 ± 0.0043	0.0607 ± 0.0199	**0.0603±0.0047**
Mean	0.1023 ± 0.0017	0.1114 ± 0.0049	0.1054 ± 0.0064	0.1735 ± 0.0027	0.0984 ± 0.0092	0.0970 ± 0.0018	0.1102 ± 0.0158	**0.0936±0.0014**

TABLE VII

TIME COST OF DIFFERENT METHODS ON THE REAL DATASETS

	$L_{1/2}-NMF$	$gtvMBO$	$SDNMF-TV$	$SSAINMF$	$HGNMF-FS$	$MAC-U$	$GL1-DNMF$	$ours$
Samson	**0.0712**	0.5307	45.9821	0.1667	4.8252	58.2871	20.7178	18.925
Urban	**0.4619**	7.4338	551.2089	0.6396	153.4422	349.9723	255.2502	248.0185
Cuprite	0.7562	4.3206	717.0183	**0.5803**	85.5194	399.5541	202.678	193.7043

of roof #1 well, specifically, for the estimated map by the method SDNMF–TV, the endmember of asphalt appeared in the map of roof #1. The methods of SSAINMF, HGNMF–FS, and ours recovered the maps better than the other methods; however, SSAINMF and HGNMF–FS, respectively, failed to recover the map of asphalt, dirt. Additionally, the results in Table V show that the proposed method MGNMWF can extract most of the endmembers accurately, especially, the proposed method MGNMWF obtained the mean value of SAD is better than the other methods. Compared to the deviations of other algorithms, our proposed algorithm is significantly more accurate. All in all, the proposed method MGNMWF outperforms the other methods not only in estimating the abundances but also in extracting the endmembers.

3) Cuprite: The Cuprite dataset is a representative dataset for the study of unmixing. It was acquired in 1997 using an airborne visible/infrared imaging spectrometer. The imaging region is located in a mining area in Nevada, USA. A 250×191 subimage containing 12 endmembers is experimented with. Since the real ground truth is absent, the maps at[2] are usually used as the reference maps. Moreover, this dataset contains 224 bands in the range of [0.370–2.480] μm. Before unmixing, the contaminated bands 1–2, 104–113, 148–167, and 221–224 are removed, and the remaining 188 bands are employed for the experiments. Compared to the synthetic datasets and the Urban datasets, the Cuprite dataset is highly mixed. The number of superpixels is still set as 100, but the parameter λ is increased and set as 0.01; correspondingly, ρ is increased and set as 0.005. The quantitative results and the estimated abundance maps are shown in Table VI and Fig. 14, respectively. As shown in Table VI, the values of SAD clearly demonstrate that the proposed method MGNMWF accurately extracted most of the endmembers compared to the other methods. Additionally, MGNMWF is more stable than other algorithms. Specifically, the mean value of our method is better than the other methods, which indicates that the overall performance of our proposed method, MGNMWF, in extracting the endmembers is better than the other methods. Additionally, Fig. 14 clearly shows that the estimated abundance maps by the proposed MGNMWF method align well with the reference abundance maps.[2] Therefore, the advocated spectral similarity regularizer and the superpixel-guided higher-order graph are highly helpful for the unmixing task.

4) Time Complexity: To analyze the time complexity, we show the time cost of different methods on the real datasets in Table VII. The results in Table VII show that the cost time of the proposed method MGNMWF is not better than the methods of $L_{1/2}$–NMF, gtvMBO, SSAINMF, and HGNMF–FS. Compared to the methods of $L_{1/2}$–NMF and gtvMBO, the proposed method MGNMWF spends time capturing the relationship between the spectra and the higher-order relationship between the pixels in each superpixel; as a result, the time cost of the proposed method MGNMWF on different real datasets is higher than the methods of $L_{1/2}$–NMF and gtvMBO. In contrast to the proposed method MGNMWF, the method SSAINMF only computes the relationship between the pixels in the superpixel and the central seed pixel, but

[2]https://speclab.cr.usgs.gov/cuprite95.tgif.2.2um_map.gif

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without computing the spectral similarity, and the method HGNMF–FS captures the global relationship of all the pixels. Therefore, the time cost of the proposed method MGNMWF on different datasets is higher than the methods $L_{1/2}$–NMF, gtvMBO, SSAINMF, HGNMF–FS. Fortunately, the cost time of the proposed method is lower than that of the SDNMF–TV, MAC–U, and GL1–DNMF methods.

VI. Conclusion

In this article, to solve the problem of accurately separating the adjacent pixels at the boundary, a spectral–spatial coupling segmentation method is advocated to guide the local higher-order graph. Specifically, the proposed method adaptively generates superpixels using edge information to enhance the homogeneity of the neighborhood. Then, a higher-order graph is established on the superpixels to explore the higher-order relationships of pixels within the neighborhood. Additionally, we overcome the shortcomings of the existing weighted factorization by considering the spectral similarity. Results on synthetic and real datasets demonstrate that the proposed method outperforms other recent methods. Unfortunately, the time cost of the proposed method, MGNMWF, is not the fastest. In the future, we will develop a more efficient method that explores pixel-deep neighborhoods using advanced semantic similarity. This will further strengthen our method's ability to handle complex nonlinearity.

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